

CHEM 101 Topic 9: Stoichiometry Calculations — Answer Key

Section 1: Mole-Mole Calculations

Problem 1: 8.50 mol NH_3 .

Problem 2: 9.00 mol H_2 .

Problem 3: 1.88 mol O_2 .

Section 2: Mole-Mass Calculations

Problem 4: 12.1 g NH_3 .

Problem 5: 27.9 g Fe.

Problem 6: 245 g H_2O .

Section 3: Molar Solutions

Problem 7: 0.075 mol HCl.

Problem 8: 4.38 g NaCl.

Problem 9: 2.87 g AgCl.

Section 4: Percent Yield

Problem 10: 84.8%.

Problem 11: 44.0 g.

Problem 12: 89.7%.